

Fiscal Structure and China's Renewable Energy Buildup

A GRF proposal for departmental feedback

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The puzzle is asymmetric local implementation

Production

Local governments often build **beyond assigned targets or subsidy allocations**.

- Steel and housing: overinvestment
- Renewables: construction ran ahead of the subsidy program

Redistribution

The same governments seldom exceed mandates for **welfare, education, or health**.

- Rigid obligations
- Limited local initiative beyond the mandate

Why does one institutional structure generate such different responses?

Why do Chinese local governments respond more strongly to **production incentives** than to **redistribution mandates**?

Fiscal dependence

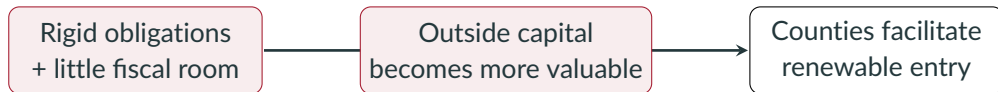
Constrained counties value externally financed production, while redistribution creates continuing fiscal obligations.

Promotion tournaments

Career-minded officials respond more strongly to targets, deadlines, and promotion incentives.

Core test: Does fiscal dependence predict renewable construction more strongly than county receipts of formula-allocated education and health transfers?

Fiscal pressure raises the value of outside investment



Motive

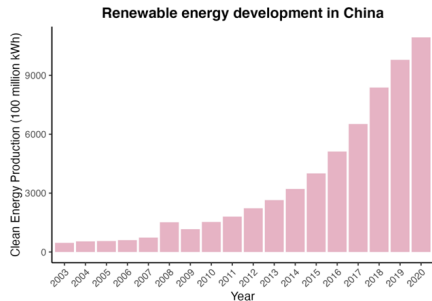
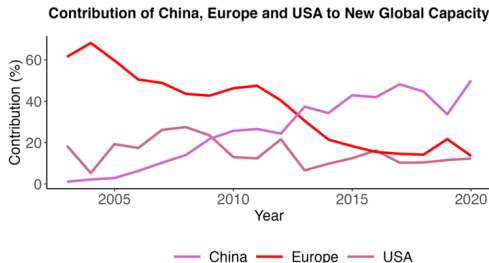
Central support makes projects viable and firms finance most investment, making renewable entry fiscally attractive.

Capacity

Counties can accelerate entry through land, permits, planning, and grid coordination.

Prediction: fiscal constraints matter most when external conditions make projects viable.

Why renewable energy? Common incentives, local discretion



- Broad geographic eligibility
- Common FIT schedule and deadlines
- County control over land and permits
- Construction independently visible by satellite

Solar is the primary test; wind provides a cross-technology replication.

Pilot evidence supports the fiscal premise—but not a simple subsidy story

24,870

county-years, 2010–2020

+3 ha

annual solar area per ¥1,000
increase in per-capita gap
(county of 500,000)

12%

of mean annual addition

- The relationship remains positive with province–year fixed effects.
- It becomes stronger when global solar costs fall.
- The fiscal-gap $\times$ FIT-premium interaction is statistically insignificant.

Interpretation: lower technology costs, rather than the FIT premium, appear to activate the relationship.

Budget tables unpack the fiscal gap

交口县二〇二六年一般公共预算收入汇总表

项 目	金额	备注
一般公共预算收入总计	176709	
一、一般公共预算收入	96590	
（一）专项收入	2050	
其中：教育费附加收入	1100	
地方教育附加收入	590	
残疾人就业保障金收入	450	
（二）一般性财力	94450	
二、上拨补贴收入	52209	
（一）返还性收入	-1082	
1、所得税基数返还	36	
2、成品油价格和税费改革返还收入	243	
其中：成品油价格改革返还基数	32	
退管所下划基数	136	
退管所下划基数	15	
2015年原车船税补助公安交警经费	60	
3、增值税返还	4222	
4、增值税五五分享所得税收入	-5583	
（二）一般性转移支付收入	52500	
1、均衡性转移支付补助收入	17807	
其中：下达2016年省对市县均衡性转移支付增量资金	868	
提前下达2016年省对市县调整工资等一般性转移支付	954	
提前下达2016年社区事务转移支付资金	38	
提前下达2016年乡镇工作补贴转移支付资金	1073	
下达调整工资政策均衡性转移支付	1135	
提前下达2016年农业转移人口市民化奖励资金	44	
下达鄂湘两省交界台办的补助资金	252	
提前下达2016年绿色农业转移支付	1245	

Jiaokou County, 2026 general public budget revenue

This table distinguishes

Own fiscal capacity RMB 944.5m

Higher-level subsidy revenue RMB 522.09m

of which: transfer payments RMB 525m

green-development transfer RMB 12.45m

What the GRF adds

Pre-entry coding of these revenues alongside payroll, mandated programs, debt service, and other rigid spending.

Which constraint predicts project entry—not merely whether a fiscal gap exists?

One county-year panel tests three explanations

	Outcome	Main test
P1	New solar capacity	FIT premium and deadlines versus technology costs
P2	New capacity and subsidy-list gap	Lagged fiscal space $\times$ FIT premium / costs
P3	Same outcomes	Career clocks: promotion window and tenure

- County and year fixed effects; province-clustered and wild-bootstrap inference
- Event studies around tariff deadlines, the 2018 halt, and grid parity
- Formula-allocated education and health transfers as placebo outcomes

The project has four linked objectives

1. **Fiscal mechanism:** Test whether rigid obligations increase demand for externally financed construction and whether subsidies or falling costs activate it.
2. **Rival explanation:** Compare budget structure and career incentives in the same models, with social-transfer placebos.
3. **Public data:** Release a new budget classification linked to energy projects, satellite construction, officials, and lobbying.
4. **Comparative scope:** Identify which features of China's industrial mobilization travel to other fiscal systems.

The project builds on a substantial existing data platform

新能通云
NEW ENERGY CLOUD

Electricity Price Subsidy > Subsidy List Announcement

Electricity Price Subsidy

当前位置: 电价补贴 > 补贴清单公告

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集中式 非自然人分布式

Centralized (photovoltaic) Non-natural person distributed (photovoltaic)

项目名称:

项目地址:

能源类型:

项目类型:

并网规模 (MW):

上网电价 (含税) (元/千瓦时):

纳入目录时间:

全部机组并网时间:

纳入目录批次:

Project List

项目列表:

Serial Number	Project Name	Detailed Address	Energy Type	Project Type	Project Company	Grid-Connected Capacity (MW)	Full Unit Grid-Connected Time	On-Grid Electricity Price (Tax-Included) (Yuan/kWh)	Year Listed in Scale Management	Business Type	Time Included in Catalog	Batch Included in Catalog
序号	项目名称	详细地址	能源类型	项目类型	项目公司	并网规模 (MW)	全部机组并网时间	上网电价 (含税) (元/千瓦时)	列入规模管理年份(年)	业务类型	纳入目录时间	纳入目录批次
1	青岛润莱风...	山东省青岛...	风电	陆上集中式...	青岛润莱风...	20	2020-12-30	0.6	2017	申报	2024-08-27	2024年补...
2	武城县三自...	山东省德州...	风电	陆上集中式...	武城县三自...	50	2020-12-12	0.57	2018	申报	2024-08-27	2024年补...
3	华能新疆吉...	新疆维吾尔自治区	太阳能发电	集中式光伏	华能新疆吉...	100	2020-12-21	0.4	2020	申报	2024-08-27	2024年补...
4	中电建中卫...	宁夏回族自治区	风电	陆上集中式...	中卫市美峰...	182	2020-12-29	0.45	2018	申报	2024-08-27	2024年补...
5	龙潭天津青...	天津市市辖区	风电	陆上集中式...	天津龙潭青...	37.4	2020-12-13	0.6	2015	变更	2024-08-27	2024年补...
6	林源大成木...	新疆维吾尔自治区	风电	陆上集中式...	木垒县林源...	50	2020-12-30	0.49	2016	申报	2024-08-27	2024年补...
7	浦城鑫源电...	福建省南平市	太阳能发电	集中式光伏	福建省浦城...	19.38	2019-06-30	0.7	2016	申报	2024-08-27	2024年补...
8	华电兴化沙...	江苏省泰州...	风电	陆上集中式...	华电兴化太...	50	2020-12-28	0.6	2017	申报	2024-08-27	2024年补...

- **In hand:** 8,000+ subsidized solar and wind projects
- **In hand:** satellite solar footprints through 2020; extension to 2025 planned
- **Validated:** budget disclosures contain the required components; national collection is proposed
- **In hand / planned:** official biographies and coal-plant data; approximately 30 interviews

Validation: automated extraction, hand-coded audits, and fiscal-statistics reconciliation

Pathways to impact: evidence, engagement, and open data

Beneficiaries	What they gain	Route to impact
International organizations and governments	Evidence for assessing rapid investment cycles, grid pressure, and the respective roles of subsidies, costs, and local fiscal conditions	Policy briefs, media commentary, and conference engagement
Developing-country policymakers	A clearer account of which elements of China's green industrial policy are transferable under different fiscal and land institutions	Executive education, workshops, and a teaching module
Researchers, analysts, and energy-industry actors	Reusable data and a framework for anticipating investment around tariff changes, cost declines, and approval windows	Public data release with documented measures and replication materials

Impact will be tracked through engagement feedback, data downloads, and citations.

Previous panel comments: positive and negative

Positive comments

- A timely and important energy-transition question
- A granular, multi-source dataset as the main innovation
- Substantial preliminary work, access, and a feasible timetable
- Strong scholarly and policy-impact potential

Negative comments

- An insufficiently sharp theoretical and literature contribution
- Too many components and units without one clearly articulated framework
- Under-specified identification, measurement, and mixed-method integration
- Limited demonstration of sector expertise and evolving policy dynamics

The task now is to turn the data advantage into one focused, testable argument.

Why me? Relevant theory, data, and sector access

1. **The project extends my research agenda.**

My published county-level work examines redistribution under economic pressure and develops a framework for comparing central–local implementation.

2. **The energy-specific groundwork is already underway.**

A companion paper assembles the National Grid project catalog, satellite construction, liaison-office budgets, and regulator biographies.

3. **I bring relevant sector access.**

My earlier work as an energy-sector journalist supports interviews with industry experts, county finance officials, and energy-SOE managers.

The GRF integrates these strands in one county-year research program.

How should I position this proposal for RGC reviewers?

1. **Positioning:** Which framing best conveys the project's contribution and its novelty beyond the companion paper?
2. **Investigator fit:** How can I present myself more convincingly as the right person to carry out this project?
3. **Scope:** What should I cut or simplify to make the project look focused and feasible within one GRF?
4. **Scheme:** Given the project's scope and my career stage, should I submit it to ECS or GRF?

I would especially value advice on **what a reviewer should remember**.